4future.tech

Technology Basics

Local, Realtime, Voice AI

No-Cloud, Austrian-German, Interactive.
Technology-Intro + Current Solution

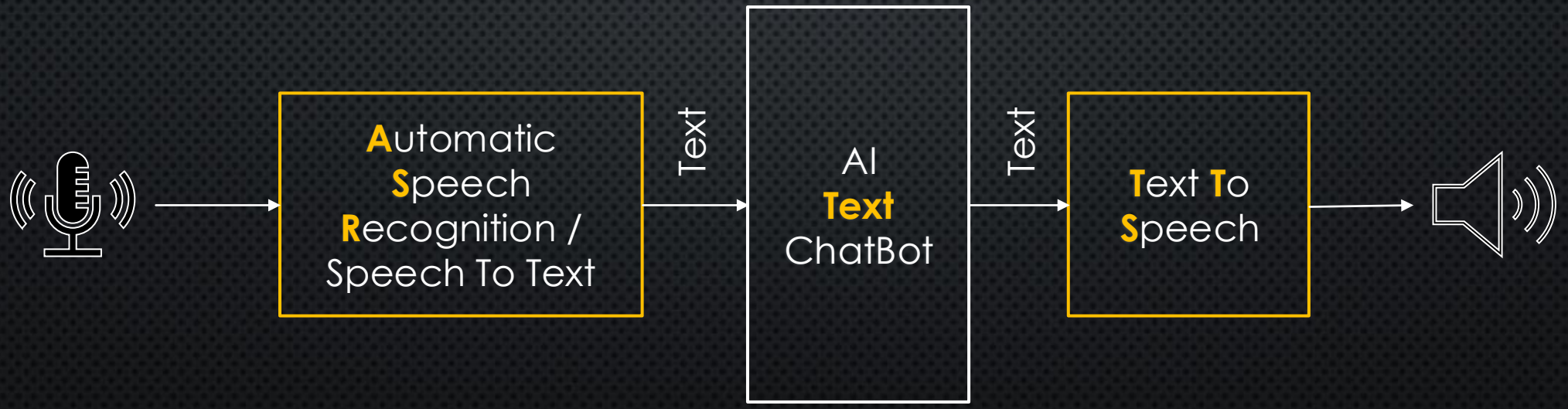


Andi Kunar
andreas@kunar.at

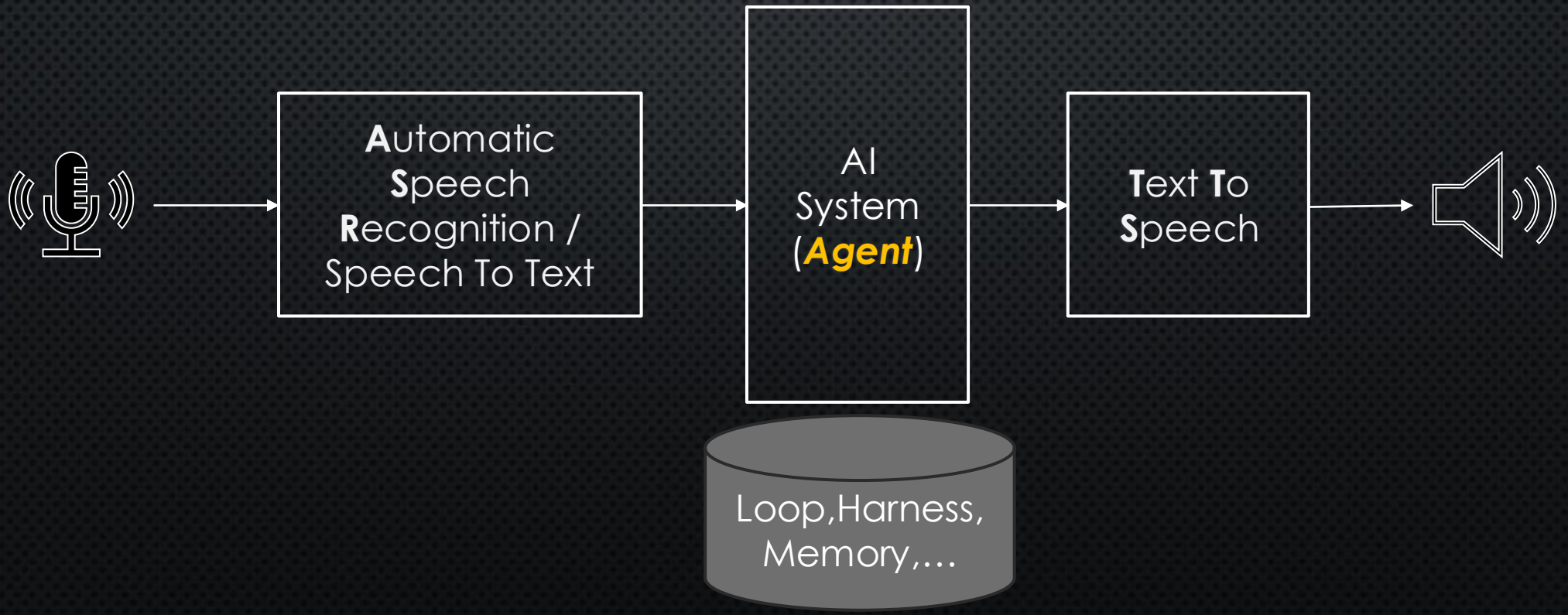
REAL-TIME VOICE AI CHATBOTS



LOCAL VOICE AI SHOULD BE EASY …

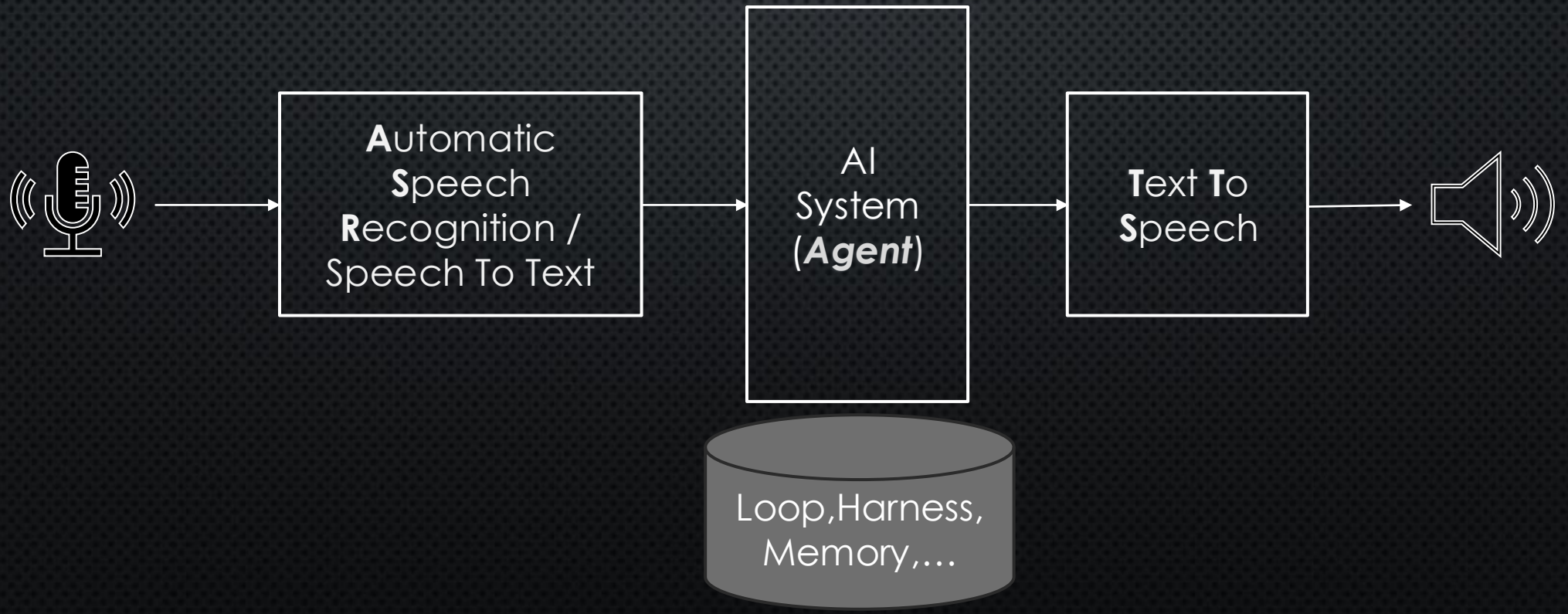


VOICE AI ISN'T THAT EASY



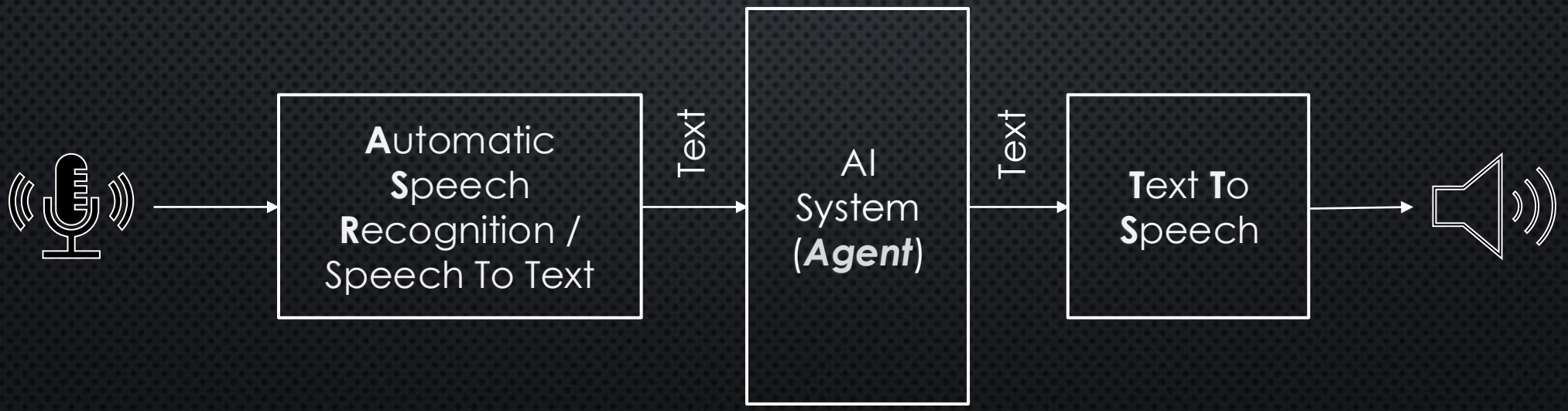
VOICE AI ISN'T EASY AT ALL

PIPELINE (ONE AFTER THE OTHER)



REAL-TIME VOICE AI REALLY ISN'T EASY

PIPELINE



webRTC openAI realtime APIs openAI APIs (streaming) openAI realtime APIs

my recommendation: mix & match via standard APIs

LOCAL

BENCHMARK ASR/TTS HARDWARE

E.G. CHEAPEST APPLE M5 + 16GB RAM

… IT JUST NEEDS TO BE FASTER THAN REAL-TIME

AUSTRIAN-GERMAN



TEXT-TO-SPEECH

(1.2-1.5X REAL-TIME)

LOCAL CLONING



TEXT-TO-SPEECH

CLONING & AUSTRIAN-GERMAN ??



TEXT-TO-SPEECH

REAL-TIME (STREAMING) = LATENCY

1ST TOKEN → 1ST SOUND (START-UP!)
CURRENTLY ~2S (SLOWER THAN HUMAN)

MY CHOICE: VOXTRAL-4B-TTS-2603

AUTOMATIC SPEECH RECOGNITION

SPEED: > 5-10X REAL-TIME

STREAMING/REALTIME: ½-1 S LATENCY
(SLOW HUMAN BUT MORE ERRORS)

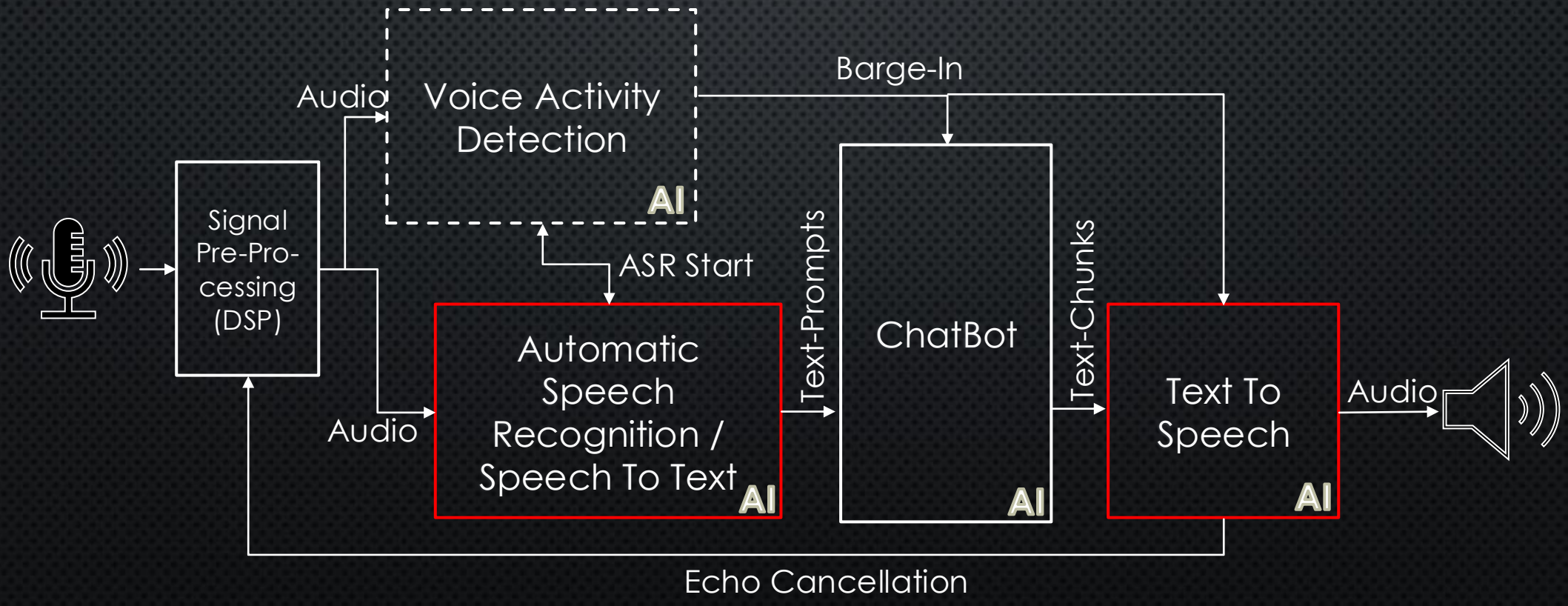
MY CHOICE: CANARY-1B-V2

THINGS GO WRONG !

- Recognition Errors (ASR **W**ord **E**rror **R**ate – human: 2-5%)
- Hallucinations (Most Noticeable in TTS)
- Real-Time Issues
  - Real-Time underruns – LLM Thinking …
  - Accuracy (streaming WER = 2-4 x batch)
  - Start-Up Artefacts

- Interruptions (barge-in) – **ChatBots / Agents act like Walkie-Talkies**

A REAL, **SIMPLIFIED** VOICE CHATBOT



ASR: Canary, TTS: Voxtral, Orchestration: LiveKit

Recommended materials

Julia Turc / YouTube (not just voice, also quantization,…)

- Realtime voice models - https://www.youtube.com/watch?v=hyhANozV9Nw
- How far are we from "Her"? - https://www.youtube.com/watch?v=kOD9rMhn4f4
- AI voices sound entirely human. What's next? - https://www.youtube.com/watch?v=B48IB_ndEoc

4future.tech

Free event with more time: Oct 6
https://4future.community/events/



Local, Realtime, Voice AI

Q&A

Andi Kunar
andreas@kunar.at